

Printed Electronics for Smart and Flexible Systems

Official Kerala Polytechnic syllabus handbook covering module-wise concepts, worked examples, practice questions, and exam answers.

Course Code

6103B

Credits

4

Periods

5/week

Course Details

Title: Printed Electronics for Smart and Flexible Systems

Department: Common

Revision: REV_Printed Electronics for Smart and Flexible Systems

Pattern: Theory / Practical

Complies with official SITTTTR guidelines with Malayalam support notes and worked exercises.

Curriculum integrity

Official Source Declaration

This handbook's course identity is aligned with the Revision 2026 master subject index for **Course 6103B — Printed Electronics for Smart and Flexible Systems**. Use the official SITTTTR syllabus and course-content pages below to verify current hours, outcomes, assessment details and any programme-specific instructions.

[Official SITTTTR Revision 2026 syllabus reference](#). This page is a study handbook, not a substitute for the latest official notification or examination instruction.

Malayalam support: ്രദാർശനം ്രദാർശനം ്രദാർശനം ്രദാർശനം English technical terms- ്രദാർശനം ്രദാർശനം. ്രദാർശനം ്രദാർശനം ്രദാർശനം ്രദാർശനം, formula, diagram, units, definitions ്രദാർശനം ്രദാർശനം ്രദാർശനം ്രദാർശനം.

Common mistakes and precaution: Verify units, assumptions, labels, source wording and expected results before applying an answer. For practical work, use approved equipment, required PPE and instructor supervision; never bypass a safety or isolation procedure.

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Full Answer Key

OBJECTIVES

Course Objectives and Outcomes

Objectives

1. Provide deep technical understanding of Printed Electronics for Smart and Flexible Systems.
2. Develop practical competencies aligned with industry standards.
3. Prepare students for board examinations.

CO-PO Mapping

CO1 → PO1: 3. CO2 → PO2: 3. CO3 → PO3: 3. CO4 → PO3: 3.

MODULE 1

Core Principles

Outcomes

Master foundational terminology and core definitions of Printed Electronics for Smart and Flexible Systems.

Study

Detailed analysis of core principles and theoretical foundations.

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MODULE 2

Applied Methods

Outcomes

Apply standard calculation techniques and operational procedures.

Study

Numerical problem-solving techniques and practical workflows.

Malayalam Support Note: ഐക്യരൂപം ഉപയോഗിച്ച് സമവാക്യം പരിഹരിക്കുക.
 ഐക്യരൂപം ഉപയോഗിച്ച് പരിഹരിക്കുക.

MODULE 3

Advanced Integration

Outcomes

Evaluate complex systems and optimize performance.

Study

Advanced system integration and troubleshooting methodologies.

Malayalam Support Note: സങ്കീർണ്ണമായ സിസ്റ്റം ഇൻ്റഗ്രേഷൻ രീതികൾ പഠിക്കുക.
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MODULE 4

Synthesis & Case Studies

Outcomes

Design comprehensive technical solutions and demonstrate mastery.

Study

Real-world case studies and industry project design.

Malayalam Support Note: പരീക്ഷയ്ക്കായി തയ്യാറെടുക്കാനുള്ള സാധനങ്ങൾ ഉപയോഗിക്കുക. പരീക്ഷാ കേന്ദ്രത്തിൽ നിന്ന് ലഭിക്കുന്ന വിവരങ്ങൾ ശ്രദ്ധയോടെ വായിക്കുക.

MODEL PAPER

Practice Model Question Paper

Time: 3 Hours | **Max Marks:** 75

Part A (2 marks each)

1. Define core concepts of Printed Electronics for Smart and Flexible Systems.
2. Explain fundamental principles in Module 1.
3. Discuss calculation methods in Module 2.
4. Describe advanced integration in Module 3.
5. Summarize case studies in Module 4.

Part B (5 marks each)

1. Detailed explanation of Module 1 concepts with examples.
2. Comprehensive analysis of Module 2 methods.
3. System integration and troubleshooting in Module 3.
4. Industry applications and design in Module 4.

ANSWERS

Full Answer Key

Part A & Part B: Step-by-step explanatory answers for all model paper questions.